熱流觀測

- 地球在形成後即不斷地向外散熱。
- 在地表量取地球散熱的速率即成爲了解內部熱構造及熱源性質的重要方法。
- 有了熱傳導係數K和地溫梯度，便完成一個熱流觀測。
- 在陸地上,地溫梯度通常在30公尺以上的井孔中量取，岩石的K值則在實驗室中測得。
- 海上熱流的量取是靠研究船垂下一探針插入沉積物，讀取一到兩公尺深的地溫梯度。
- 目前，全球地表的熱流觀測的分佈，海洋和陸地的平均熱流值都在65毫瓦/平方公尺左右。

熱流 (Q : mW/m^2)

- 熱流(Q)爲一向量，表示每秒通過單位面積的熱量。
- 若熱量的流通是用傳導的方式，即分子靠振動將熱能傳給鄰近的分子，則熱流的定義爲

$$Q = -k \, dT / dz$$

K爲熱傳導係數，表示物質傳遞熱的能力，
T爲溫度，z代表三維正交座標。
式中的負數是當熱能流向座標的正方向時，
讓q出現負值。

Fig. 4.24 Method of measuring oceanic heat flow and recovering samples of marine sediments: (a) a coring device is lowered by cable to the sea-floor, (b) when a trigger-weight contacts the bottom, the corer falls freely, and (c) temperature measurements are made in the ocean floor and the sediment-filled corer is recovered to the surface ship.

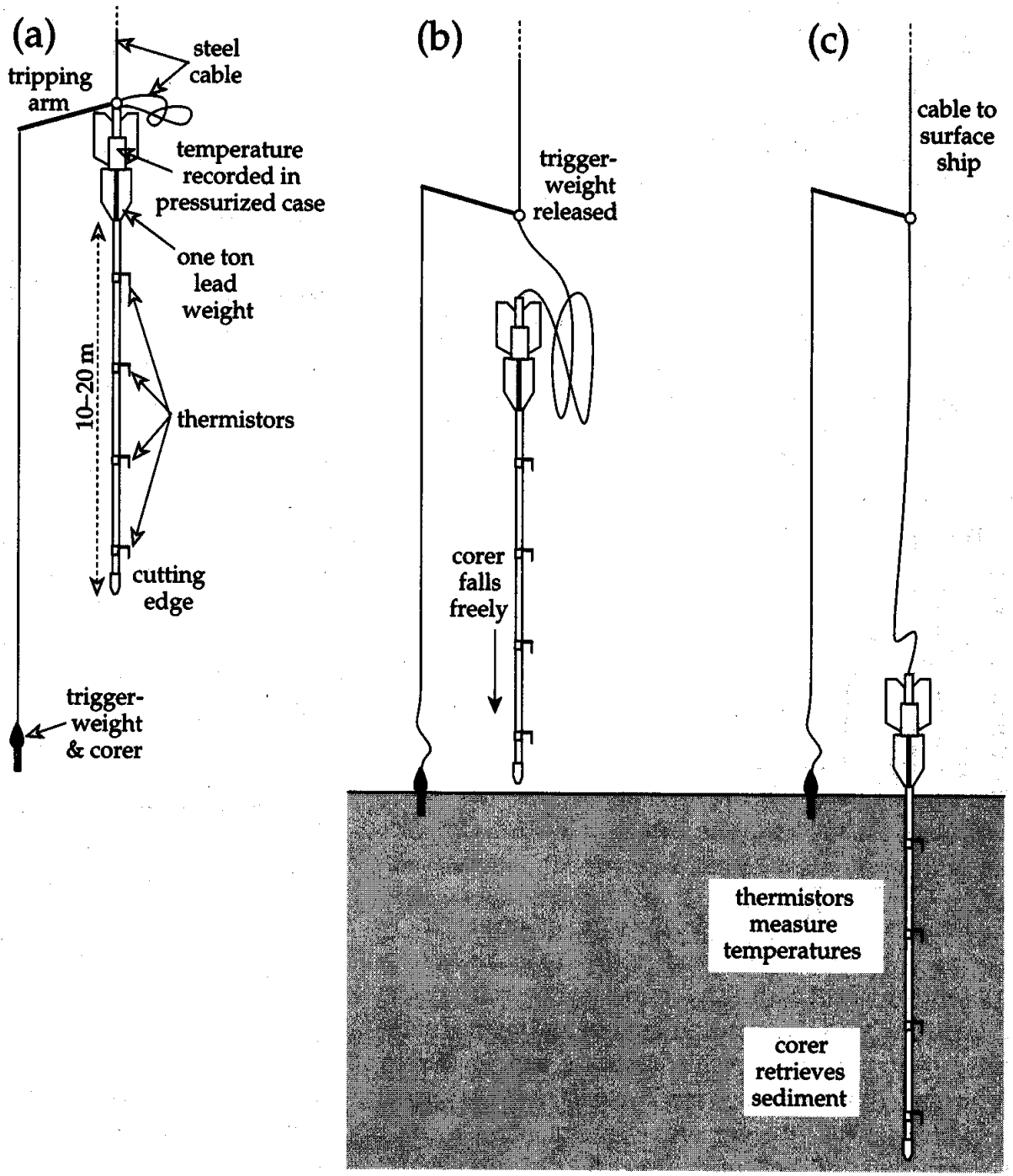


Table 4.2 *Estimates of notable contributions to the Earth's annual energy budget*

Energy source	Annual energy (joule)	Normalized (geothermal flux = 1)
Reflection and re-radiation of solar energy	5.4×10^{24}	≈4000
Geothermal flux from Earth's interior	1.4×10^{21}	1
Rotational deceleration by tidal friction	$\approx10^{20}$	≈0.1
Elastic energy in earthquakes	$\approx10^{19}$	≈0.01

熱 源

地球的熱源可歸納成兩種:

- 其一是放射性元素衰減時，由放射粒子動能轉換而來的熱能;
- 其二是地球形成時儲存下來的熱。

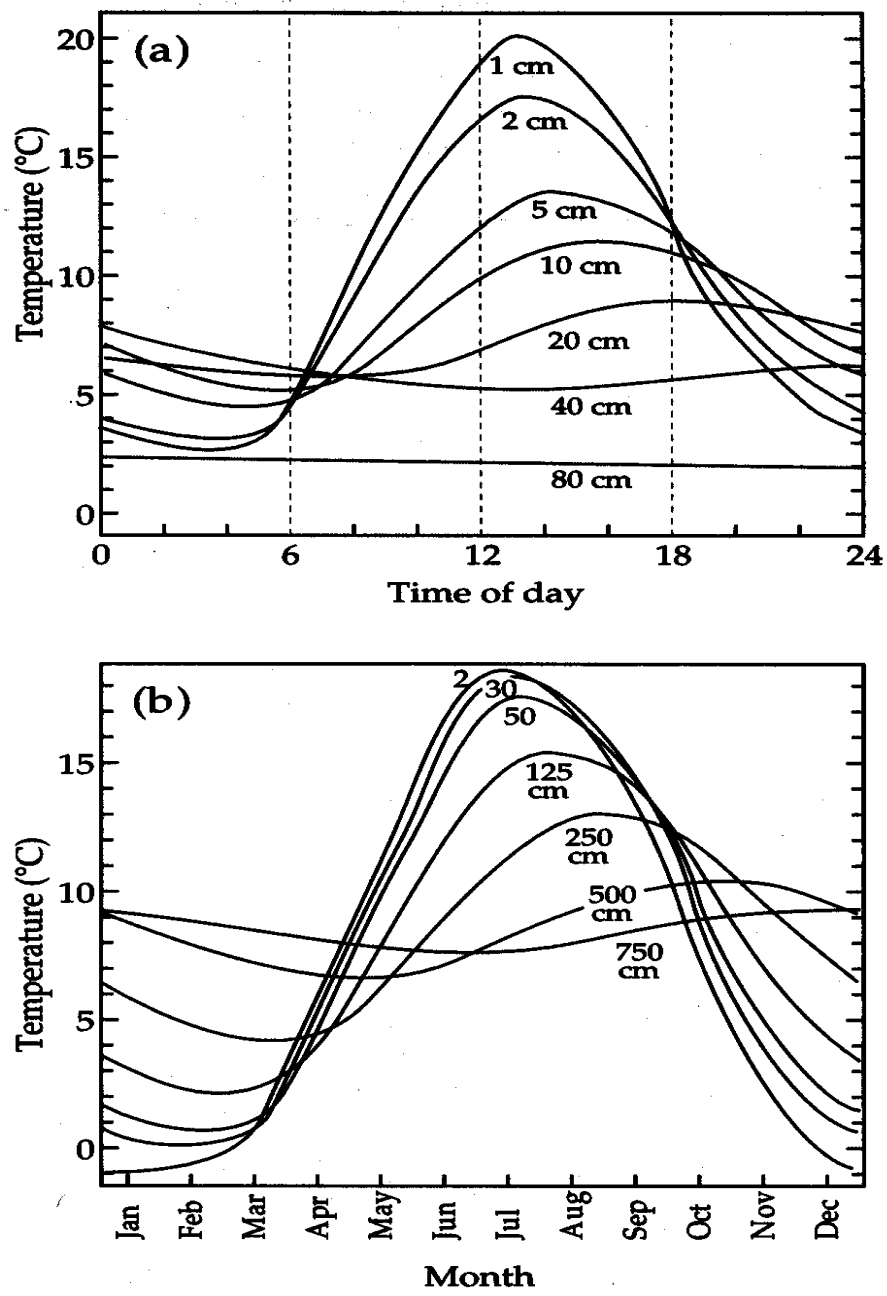


Fig. 4.18 Temperature variations at various depths in a sandy soil: (a) daily fluctuations, (b) annual (seasonal) variations.

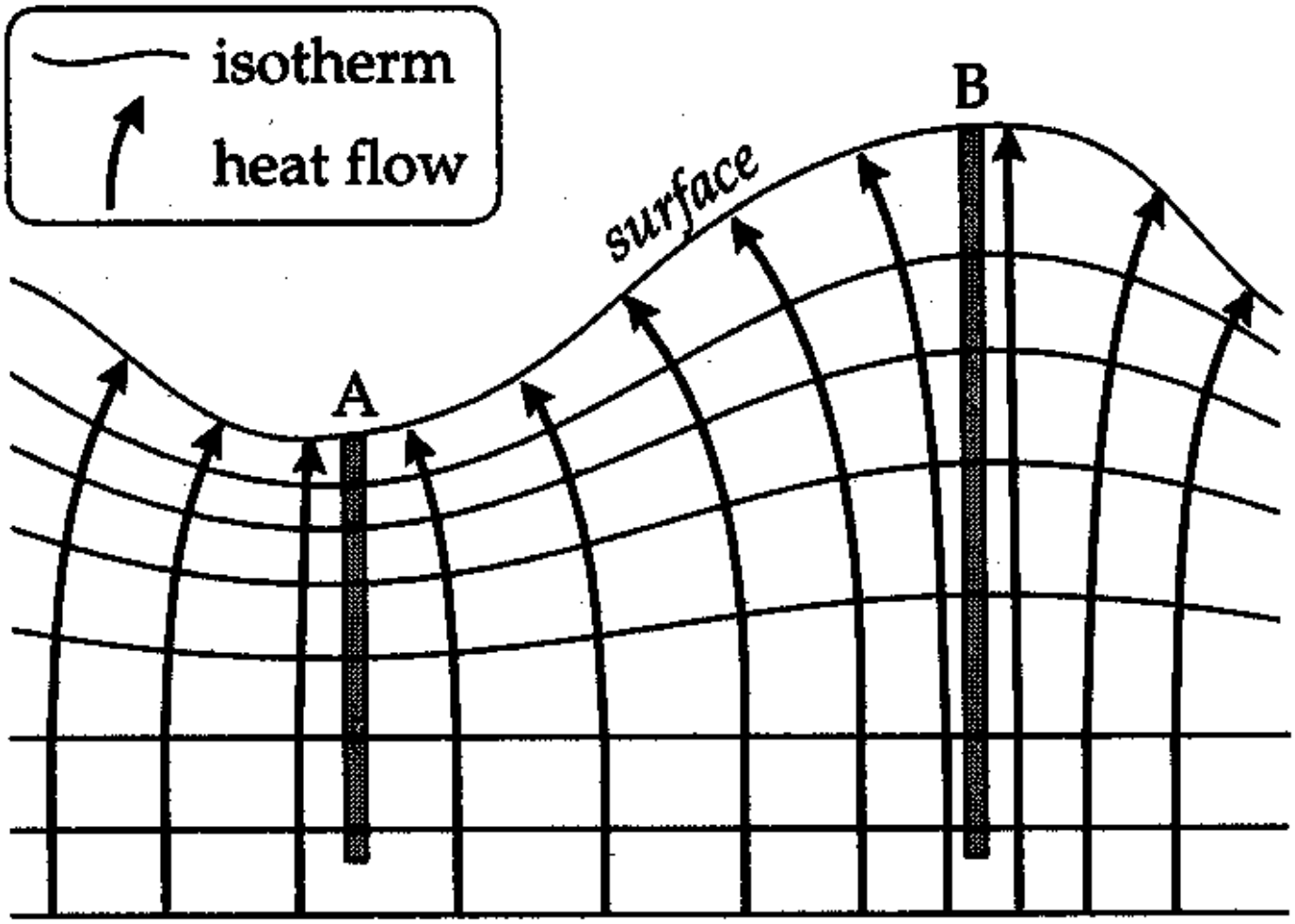


Fig. 4.21 Schematic effect of surface topography on isotherms (*thin lines*) and the direction of heat flow (*thick lines*).

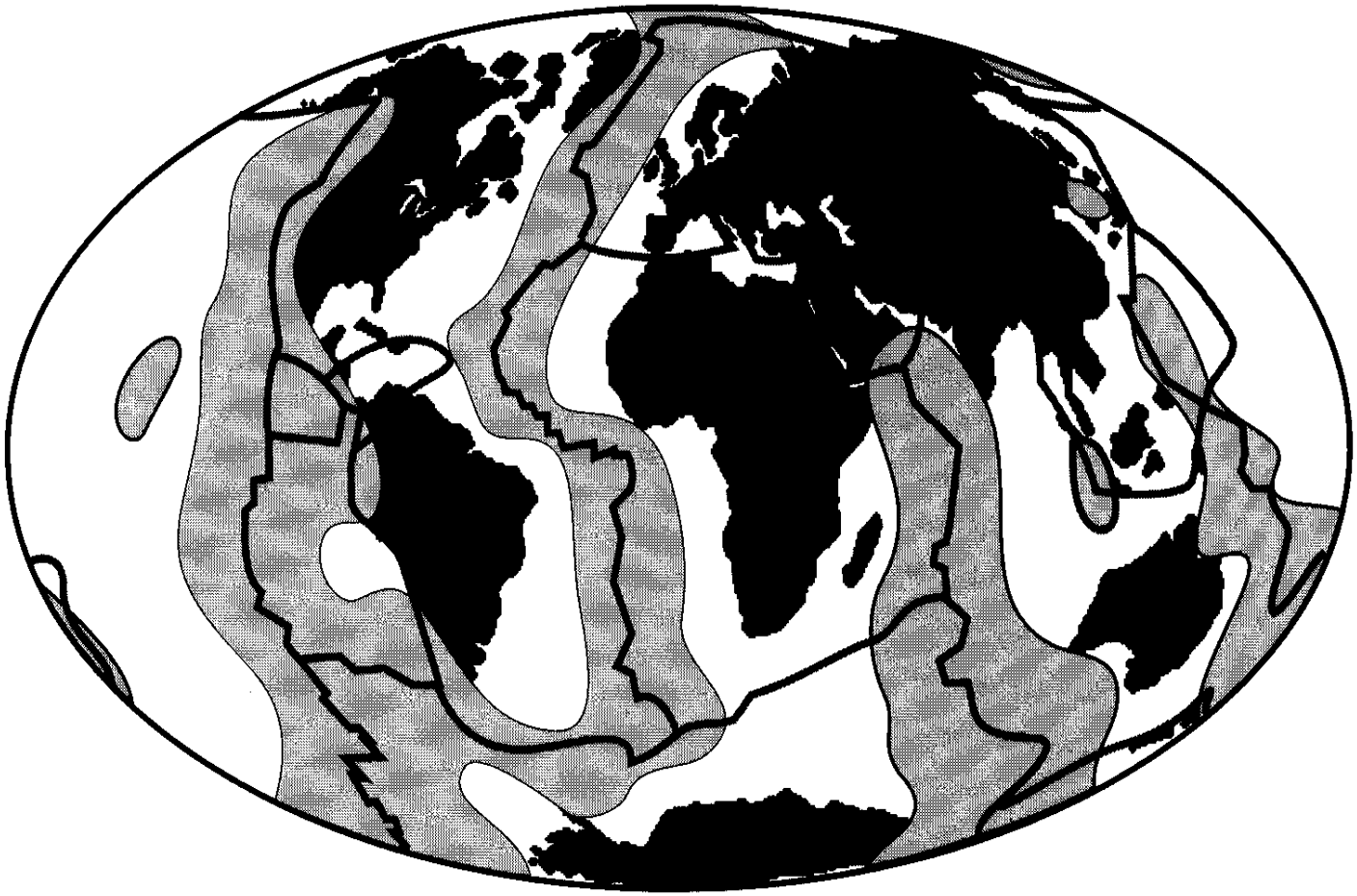


Fig. 4.29 Geographic regions where the heat flow is higher (*lighter-shaded*) and lower (*unshaded*) than the global mean heat flow; lines (*darker shaded*) mark positions of plate boundaries (after Pollack *et al.*, 1993).

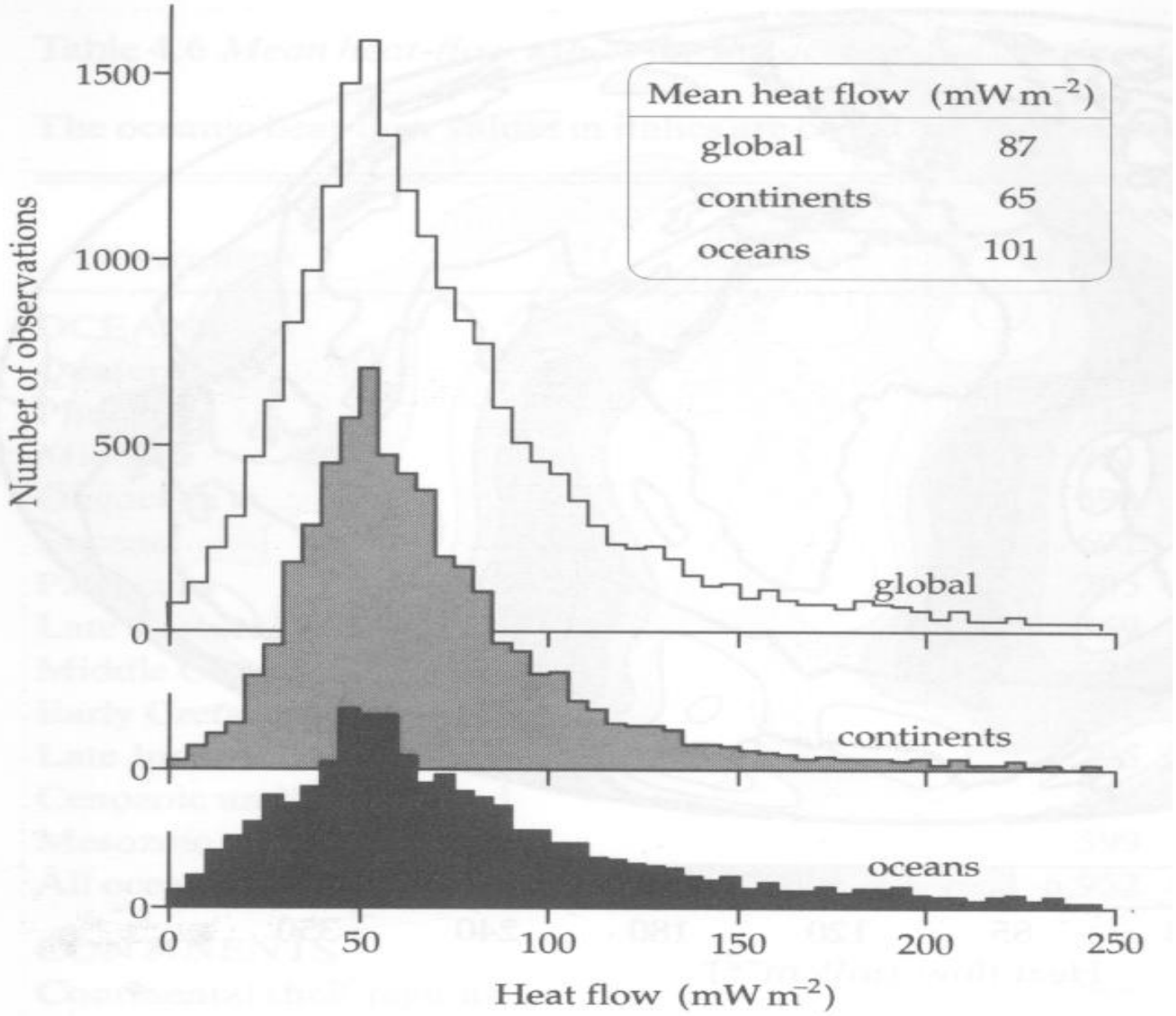


Fig. 4.27 Histograms of continental, oceanic and global heat-flow values (after Pollack *et al.*, 1993).

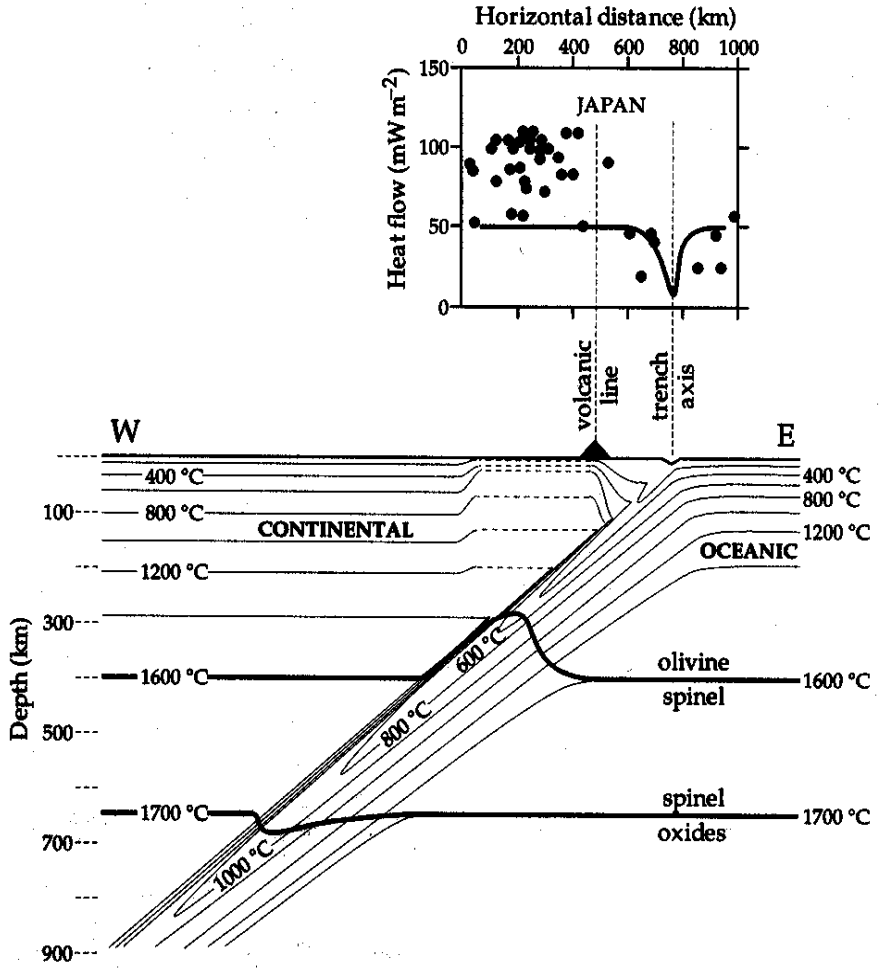


Fig. 4.33 (*Bottom*): The thermal structure of a subduction zone and back-arc region (the model of Schubert *et al.*, 1975, inverted horizontally), showing the possible isotherms in the cold subducting plate and the thermal effects of the olivine–spinel and spinel–oxide phase changes. (*Top*): comparison of heat-flow measurements across the Japanese trench with the theoretical heat flow (solid curve) computed by Toksöz *et al.* (1971).

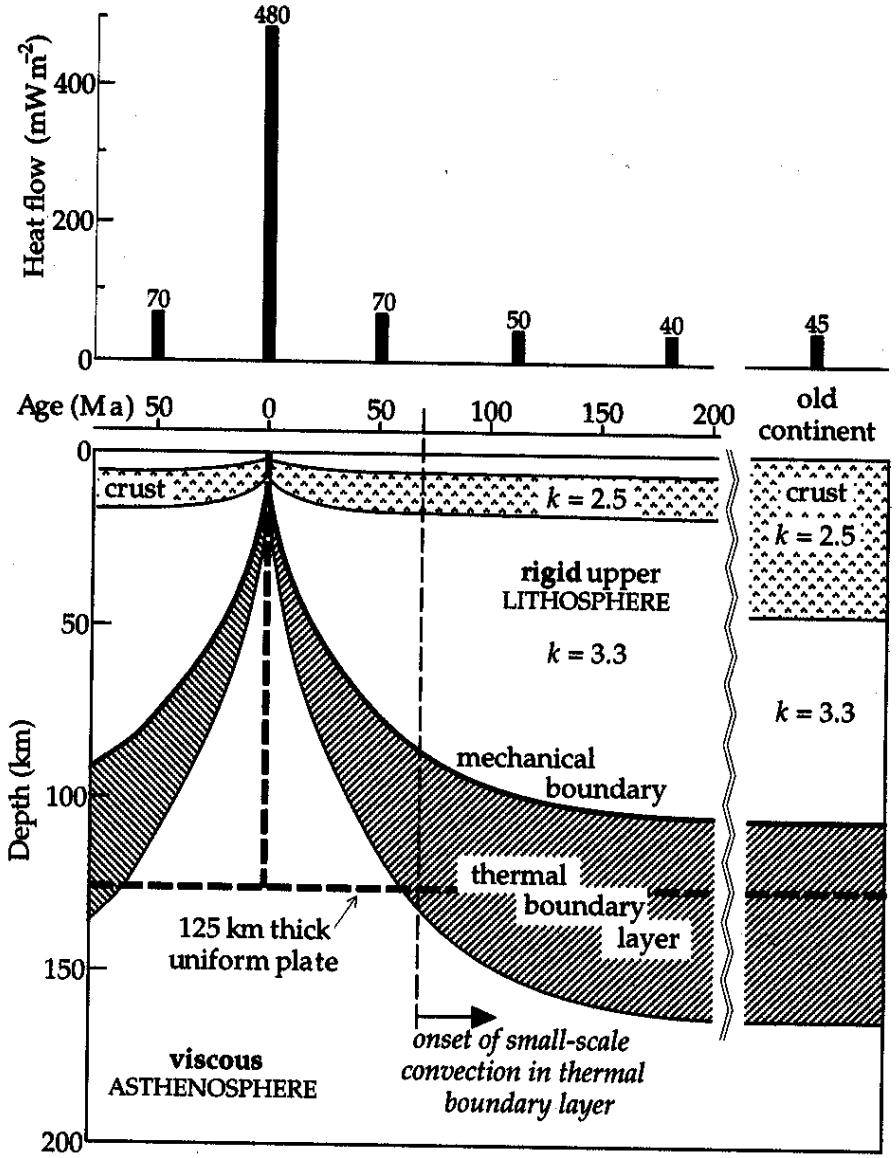


Fig. 4.32 Schematic diagram of lithospheric plate structure beneath oceans and continents. The dashed line indicates the approximation as a plate of constant thickness (based upon Parsons and McKenzie, 1978, and Sclater *et al.*, 1981).

板塊運動的力源

洋脊推力(Ridge push)
地函施力(Mantle drag)
隱沒拉力(Slab pull)
隱沒阻力(Slab resistance)
碰撞阻力(Colliding resistance)

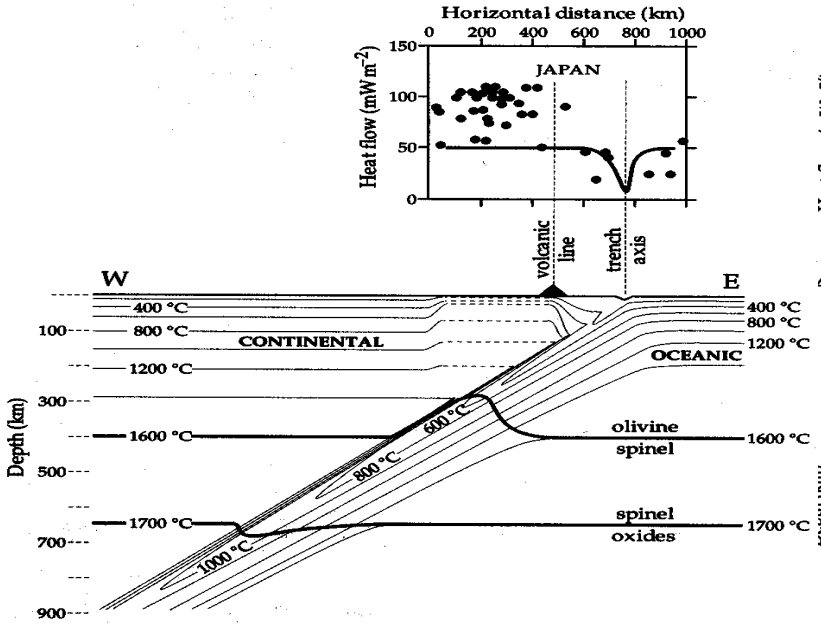
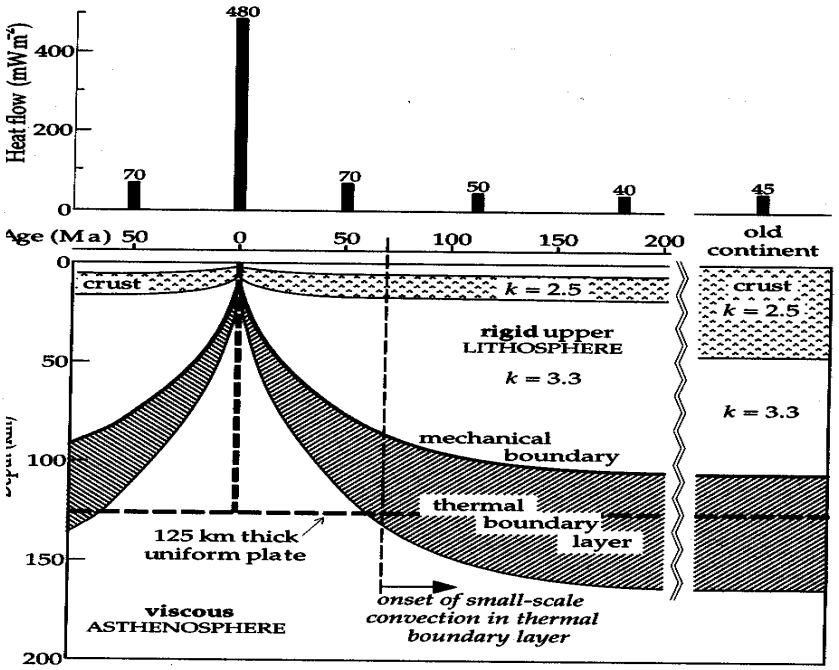


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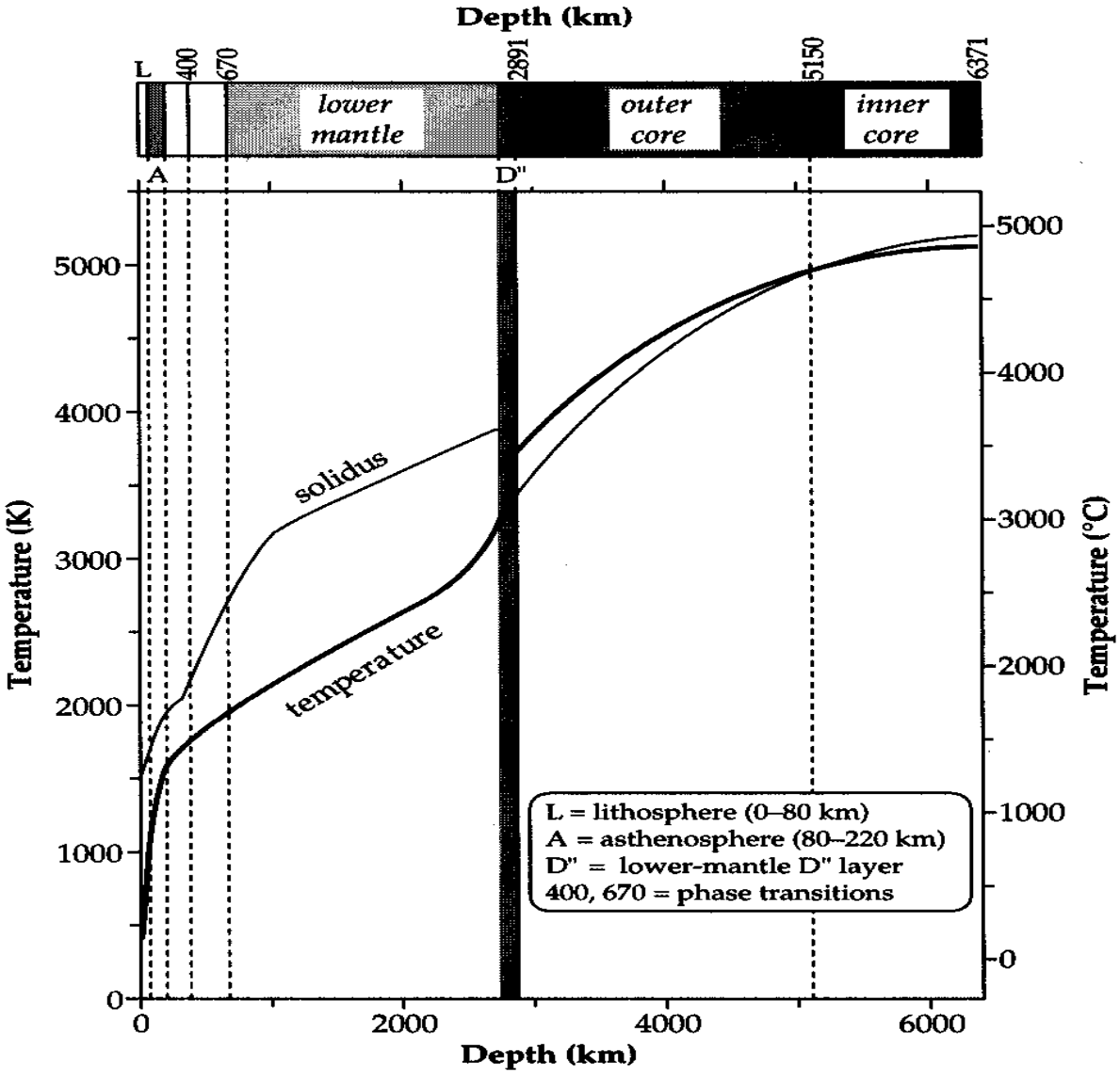


Fig. 4.14 Variations of estimated temperature and melting point with depth in the Earth (based upon data from Stacey, 1992).

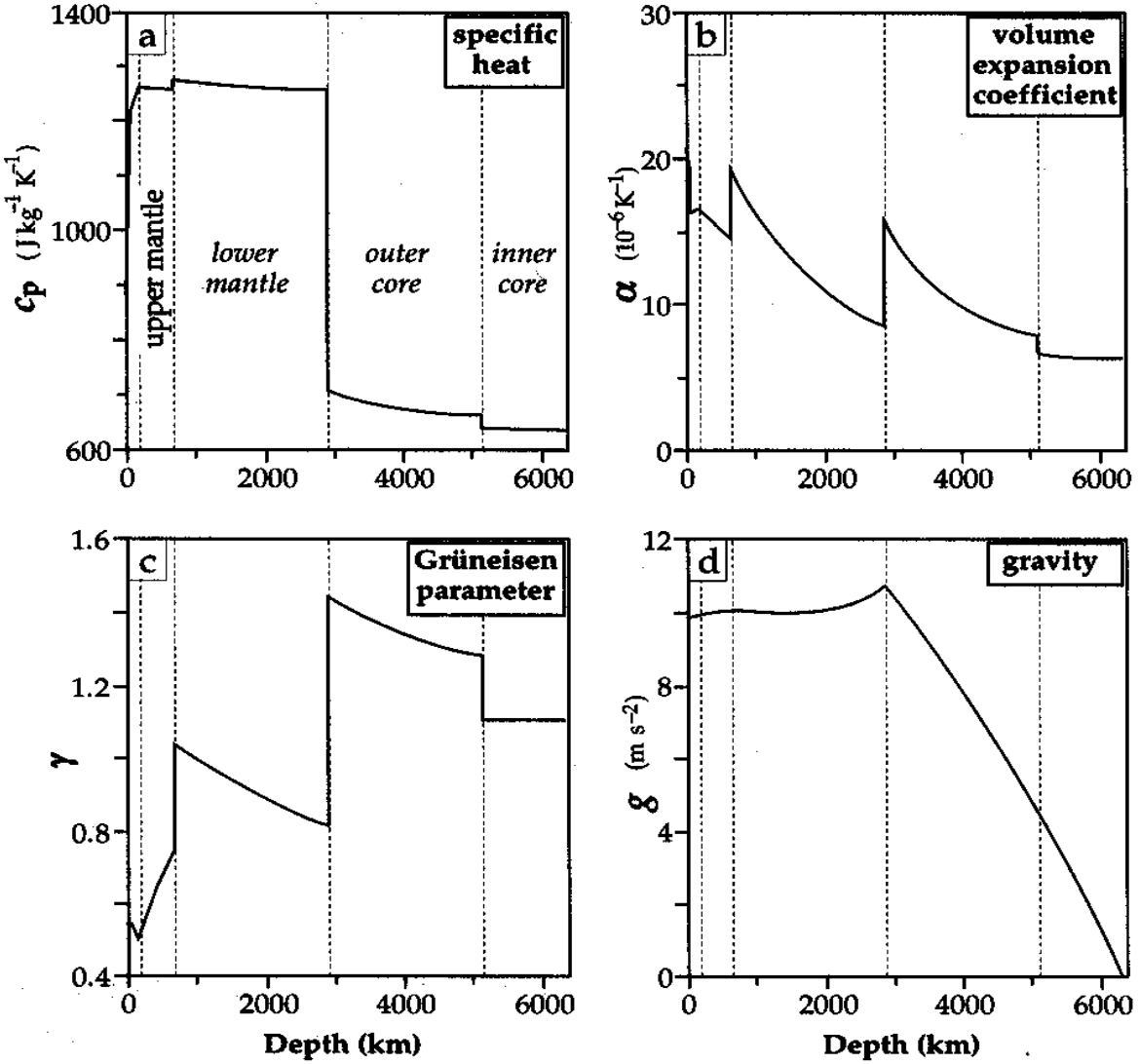
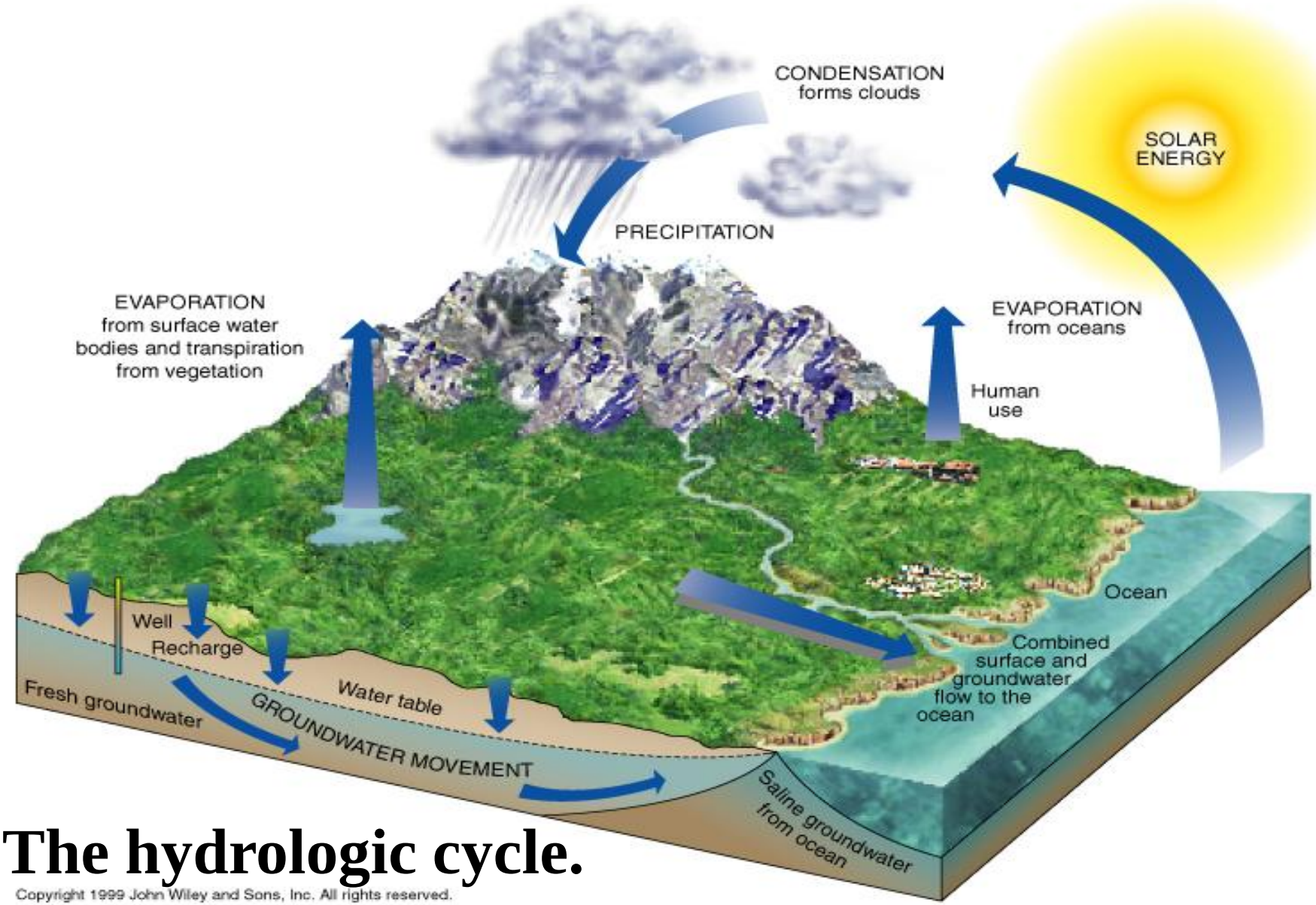


Fig. 4.13 Variations with depth in the Earth of (a) specific heat at constant pressure, (b) volume coefficient of thermal expansion, (c) Grüneisen parameter, and (d) gravity (based upon data from Stacey, 1992).

Table 4.4 *Approximate relative contributions (in %) of the main sources of heat flow in oceanic and continental lithosphere (from Bott, 1982)*

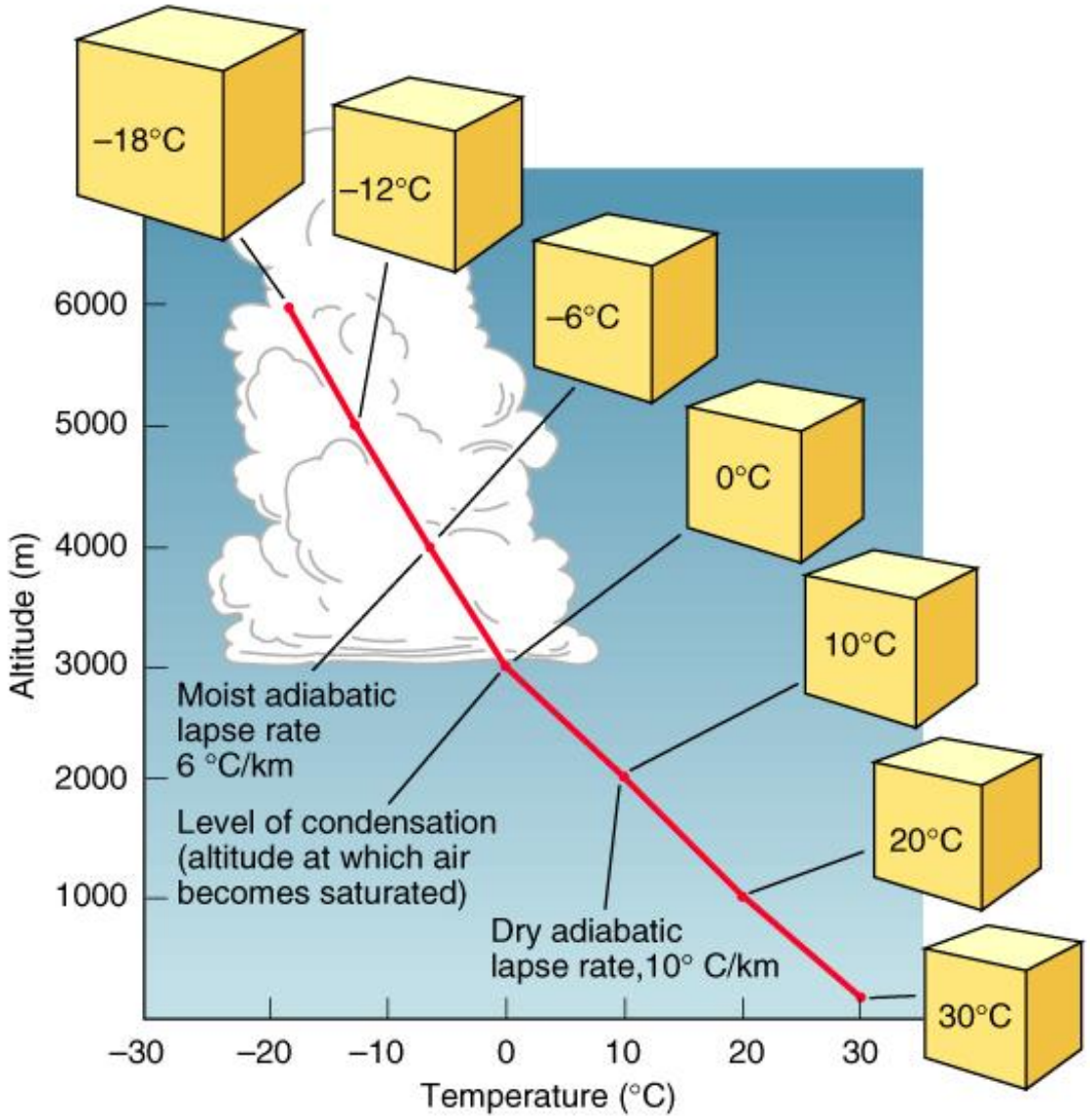
Heat source	Contribution to heat flow in: Continents (%)	Oceans (%)
Cooling of the lithosphere	20	85
Heat flow from below the lithosphere	25?	85
Radiogenic heat:	55?	5
upper crust	40	—
rest of lithosphere	15?	—

大氣中各種現象



The hydrologic cycle.

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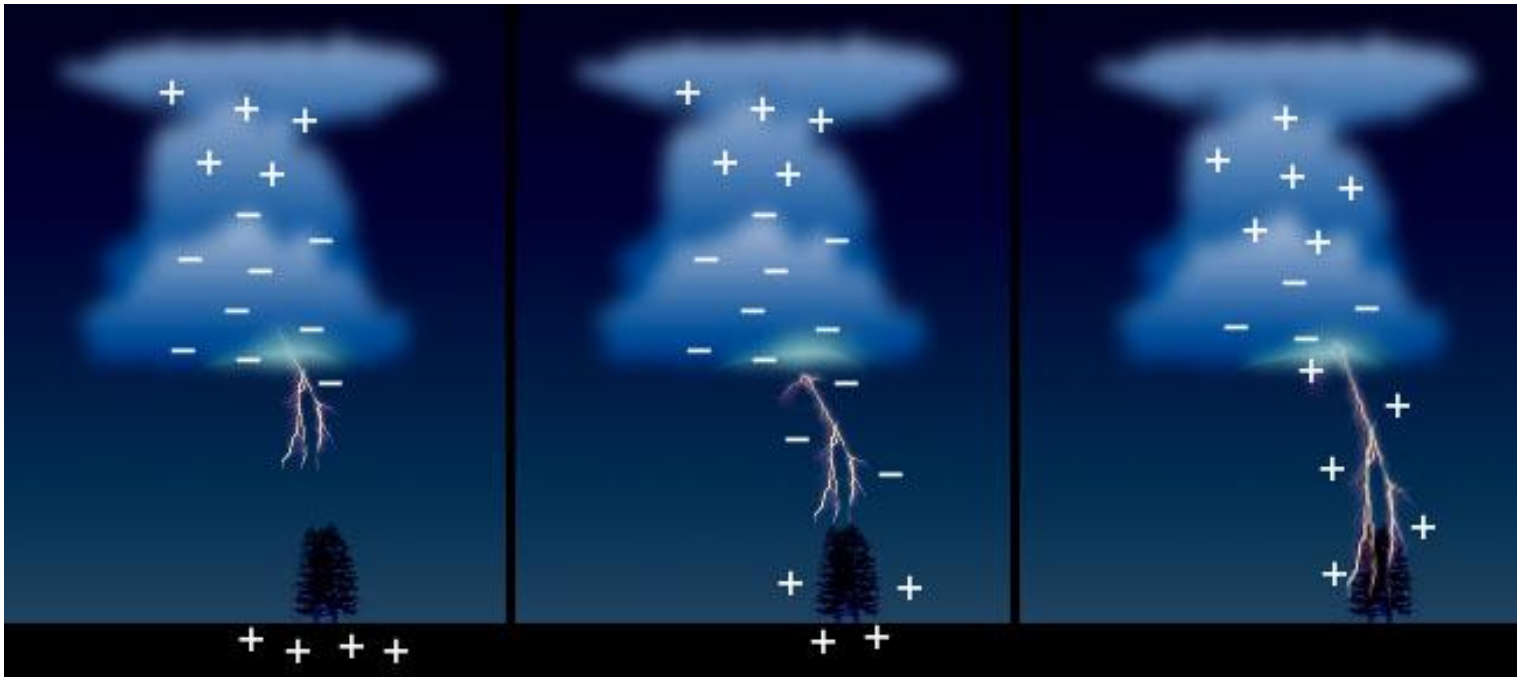


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Lightening and Thunder



根據觀測，雲層的上部帶正電，下部帶負電，因此地面便會感生了大量的正電。雲層下部的電位要比地面低得多，所以帶負電的電子便會向地面加速。於是在雲與地面之間或者在雲的不同部位之間以及不同雲塊之間激發出耀眼的閃光。這就是人們常說的閃電。



熱力學(Thermodynamics)

- 宇宙的一切變化都在熱力學(thermodynamics)的掌握之中，從一種能源改變成另一種能源，更會受到熱力學的監控。

- 熱力學規範了能量的平衡(第一定律)、能量的流動(第二定律)及能量的極限(第三定律)。

熱力學第一定律

- 根據愛因斯坦1905年提出的「狹義相對論」質量爲濃縮的能量，即 $E = MC^2$，
- 所以「熱力學第一定律」的含意實爲質能的總和不變。
- 它也包括了沒有能量變化時的「物質不滅定律」。
- 根據「熱力學第一定律」，太陽能當然可以轉換成電能，但其發電的效率卻受限於下述的「第二定律」。

熱力學第二定律: 增熵定律
熵值 (dS): dS = dQ/T

- 「熱力學第二定律」是所有物理定律的基礎，因爲它界定了一切變化的方向，包括無從捉摸的時間流向。

- 1854年R. Clausius正式提出「熱力學第二定律」，認爲在一個封閉的系統內所有的不平衡變化(自發或不可逆的變化)其「熵值」一定會持續增加直到趨於平衡爲止。
- 1865年他把能量(energy)及改變(tropae)合成Entropy這個字來說明所有可用的能量(焓，enthalpy)都會逐漸消耗並轉化成不能用的「熵」。

絕對溫度: -273℃

- 1824年N. L. Sadi Carnot在分析內燃機的熱流分佈時發現熱能無法全部用來推動引擎，其作功的效率爲 (T1-T2)/T1，

  式中的T1及T2爲進氣與出氣的溫度，

  其中浪費掉的廢熱即爲後來所孰知的「熵值」(entropy)

- 1787年J. A. Charles發現氣體受熱時，每加熱攝氏一度，其體積會膨脹約1/273。1848年Lord Kelvin以此現象倒推到低溫，他認爲氣體在-273℃時體積應爲零，因此他主張這是溫度的最低點。後來科學家乃稱此溫度爲絕對零度(0 K)，並以此點往上標定成爲絕對溫度計，這個溫度尺度即以Kelvin表示。